# 99.500 Applied Mathematics for Engineering

Homework 2 Solution

February 25, 2026

1. (a) Let $y = \frac{dC}{dx}$ and the ODE can be transformed into

$$D\frac{dy}{dx} = -r_{m_0}e^{-\lambda x} \tag{1}$$

$$\int dy = \frac{-r_{m_0}}{D}\int e^{-\lambda x}dx \tag{2}$$

$$y = \frac{r_{m_0}}{\lambda D}e^{-\lambda x} + c_1 \tag{3}$$

$$\frac{dC}{dx} = \frac{r_{m_0}}{\lambda D}e^{-\lambda x} + c_1 \Longrightarrow C(x) = -\frac{r_{m_0}}{\lambda^2 D}e^{-\lambda x} + c_1 x + c_2 \tag{4}$$

$$C(0) = 0 \Longrightarrow -\frac{r_{m_0}}{\lambda^2 D} + c_2 = 0 \Longrightarrow c_2 = \frac{r_{m_0}}{\lambda^2 D} \tag{5}$$

$$\left.\frac{dC}{dx}\right|_{x=d} = 0 \Longrightarrow \frac{r_{m_0}}{\lambda D}e^{-\lambda d} + c_1 = 0 \Longrightarrow c_1 = -\frac{r_{m_0}}{\lambda D}e^{-\lambda d} \tag{6}$$

The final expression for the monomer concentration is

$$C(x) = -\frac{r_{m_0}}{\lambda^2 D}e^{-\lambda x} - \frac{r_{m_0}}{\lambda D}e^{-\lambda d}x + \frac{r_{m_0}}{\lambda^2 D} = \frac{r_{m_0}}{\lambda^2 D}\left(1 - e^{-\lambda x} - \lambda x e^{-\lambda d}\right) \tag{7}$$

2. (a) Apply Nondimensionalization with

$$y = \frac{C_s}{C_{s_0}} \;\;\&\;\; x = \sqrt{\frac{k}{D_s}}r \tag{8}$$

$$\frac{dC_s}{dr} = C_{s_0}\frac{dy}{dx}\sqrt{\frac{k}{D_s}} \tag{9}$$

$$\frac{d^2C_s}{dr} = C_{s_0}\frac{k}{D_s}\frac{d^2y}{dx^2} \tag{10}$$

to the governing equation and boundary conditions to obtain

$$x^2y'' + xy' - x^2y = 0 \tag{11}$$

$$y'(0) = 0 \tag{12}$$

$$y(\gamma) = 1 \text{ where } \gamma = \sqrt{\frac{k}{D_s}}R \tag{13}$$

This equation (11) is modified Bessel ODE with $p = 0$ and its solution is

$$y(x) = c_1 I_0(x) + c_2 K_0(x) \tag{14}$$

With boundary condition (12) to have $c_2 = 0$ and boundary condition (13) to have $c_1 = \frac{1}{I_0(\gamma)}$. Hence,

$$y(x) = \frac{I_0(x)}{I_0(\gamma)} \Longrightarrow C_s = \frac{C_{s_0}I_0(\sqrt{\frac{k}{D_s}}r)}{I_0(\sqrt{\frac{k}{D_s}}R)} \tag{15}$$

(b) Reaction rate of absorbed oxygen ($\frac{\text{mol}}{\text{s}}$) is the flux at the surface:

$$\begin{aligned} R_0 &= -2\pi R N|_{r=R} \\ &= 2\pi R D_s \left.\frac{dC_s}{dr}\right|_{r=R} \\ &= 2\pi R C_{s_0}\sqrt{kD_s}\left.\frac{dy}{dx}\right|_{x=\gamma} \\ &= 2\pi R C_{s_0}\sqrt{kD_s}\left.\frac{I_1(x)}{I_0(\gamma)}\right|_{x=\gamma} \\ &= 2\pi R C_{s_0}\sqrt{kD_s}\frac{I_1(\sqrt{\frac{k}{D_s}}R)}{I_0(\sqrt{\frac{k}{D_s}}R)} \end{aligned} \tag{16}$$

(c) When $R \rightarrow 0$,

$$\lim_{R\to 0} R_0 \sim 2\pi R C_{s_0}\sqrt{kD_s}\frac{\frac{R}{2}\sqrt{\frac{k}{D_s}}}{1} \sim \pi R^2 C_{s_0} k \tag{17}$$

It behaves as if its entire surface is reacting with the concentration at the boundary. When $R \rightarrow \infty$,

$$\lim_{R\to\infty} R_0 \sim 2\pi R C_{s_0}\sqrt{kD_s}\frac{e^{\sqrt{\frac{k}{D_s}}R}\sqrt{2\pi\sqrt{\frac{k}{D_s}}R}}{e^{\sqrt{\frac{k}{D_s}}R}\sqrt{2\pi\sqrt{\frac{k}{D_s}}R}} \sim 2\pi R C_{s_0}\sqrt{kD_s} \tag{18}$$

$$\frac{R_0}{R} \longrightarrow 2\pi C_{s_0}\sqrt{kD_s} \text{ (gradient of a straight line when plotting } R_0 \, vs \, R) \tag{19}$$

Arbitrary values taken for generating following Matlab plot:

$$\begin{aligned} R &= 0 : 0.0001 : 1\text{cm} \\ C_{s_0} &= 2\frac{\text{mol}}{\text{cm}^2} \\ k &= 0.1\text{s}^{-1} \\ D_s &= 0.01\frac{\text{cm}^2}{\text{s}} \end{aligned}$$